

Internet Key Exchange (IKE) Protocol:

- o IKE provides framework to exchange the security parameter between two IPsec peers.
- o IKE provides a framework to exchange policies between two IPsec peers.
- o IKE is a protocol used to set up an IPsec Security Associations (SAs) security attributes.
- o Collection of parameters that the two devices will use is called **Security Associations**.
- o It provides encryption key, encryption algorithm, and mode, between IPsec peers.
- o IKE allows IPsec peers to dynamically exchange keys and negotiate IPsec SAs.
- o Using IKE, IPsec SAs can be dynamically established & removed at negotiated time, period.
- o IKE has two versions, an old version IKEv1 and a new version is an IKEv2.
- o IKEv1 was introduced around 1998 and superseded by IKEv2 in 2005.
- o IKE is a hybrid protocol made from Oakley, SKEME and **ISAKMP** protocols.
- o ISAKMP protocol is a framework for exchanging encryption keys and SA payloads.
- o Internet Key Exchange (IKE) uses UDP, Port Number **500** for sending or receiving.
- o IKE is a management protocol actually is use isakmp for key exchange.
- o IKE (Internet Key Exchange) protocols is used to establish an IPsec tunnel.
- o **IKE** is same with Internet Security Association Key Management Protocol (**ISAKMP**).
- o IKE builds the tunnels for us, but it does not authenticate or encrypt user data.

IKE Version:

- o IKE has two versions, old version IKEv1 & new version is IKEv2.
- o There are a number of differences between IKEv1 and IKEv2.
- o IKEv1 was introduced around 1998 & superseded by IKEv2 in 2005.
- o IKEv2 does not consume as much bandwidth as IKEv1.
- o IKEv2 has a built-in Keepalive mechanism for tunnels.
- o IKEv2 can detect whether tunnel is still alive while IKEv1 cannot.
- o IKEv2 supports EAP authentication while IKEv1 does not.
- o IKEv2 has built-in NAT traversal while IKEv1 does not.
- o IKEv2 to be used in mobile platforms like phones.
- o IKEv2 supports MOBIKE while IKEv1 does not.

```

> Frame 5: 206 bytes on wire (1648 bits), 206 bytes captured (1648 bits) on interface 0
> Ethernet II, Src: ca:01:03:70:00:00 (ca:01:03:70:00:00), Dst: ca:02:01:08:00:00 (ca:02:01:08:00:00)
> Internet Protocol Version 4, Src: 192.168.12.1, Dst: 192.168.23.3
> User Datagram Protocol, Src Port: 500, Dst Port: 500
4 Internet Security Association and Key Management Protocol
  Initiator SPI: c2411928cc02999f
  Responder SPI: 0000000000000000
  Next payload: Security Association (1)
  Version: 1.0
  Exchange type: Identity Protection (Main Mode) (2)
  Flags: 0x00
```

IKE Port No

Using IKE version (1.0)

We are using main mode.

Internet Key Exchange (IKE) Phases:

- o There are two phases to build an IPsec tunnel IKE phase 1 and IKE phase 2.
- o IKEv1 Phase 1 Security Associations negotiation is for protecting IKE.
- o IKEv1 Phase 2 Security Associations negotiation is protecting real user traffic.

IKEv1 Phase 1:

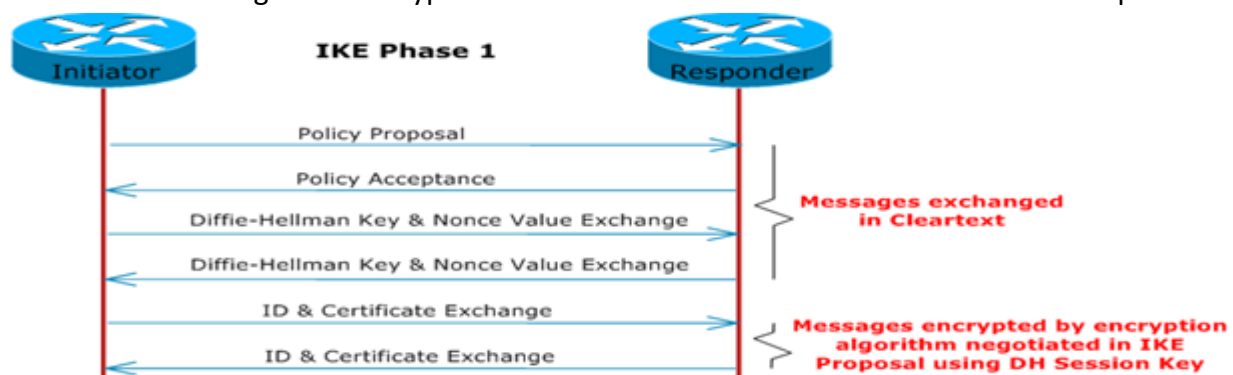
- o In IKE phase 1, two peers negotiate about the encryption, authentication & hashing.
- o In IKE Phase 1, two peers also negotiate about other protocols that they want to use.
- o In IKE Phase 1, two peers also negotiate about some other parameters that are required.
- o Main purpose of IKE phase 1 is to establish secure tunnel that can be used for IKE phase 2.
- o In Internet Key Exchange (IKE) Phase 1, an **ISAKMP** session is established.
- o Internet Key Exchange Phase 1 also called **ISAKMP tunnel** or IKE phase 1 tunnel.
- o The Internet Key Exchange (IKE) phase 1 tunnel is only use for **Management Traffic**.
- o IKE Phase 1 tunnel as a secure method to establish the second tunnel.
- o IKE phase 1 can happen in two modes main mode or aggressive mode.

IKEv1 Phase 1 Modes:

- o IKEv1 Phase 1 negotiation is in two modes, either Main Mode or Aggressive Mode.

Main Mode:

- o Main mode uses six messages while aggressive mode only uses three messages.
- o Main mode has three pairs of messages total six messages between IPsec peers.
- o The main purpose of IKEv1 Phase 1 is to establish IKE Security Associations.
- o IKE Main mode is considered more secure then aggressive mode.
- o The initiator peer that wants to build the tunnel will send the first message.
- o Initiator send own proposal to responder for the security association.
- o When the responder receives the first message from the initiator, it will reply.
- o This message is used to inform the initiator that we agree upon the attributes.
- o This way the responder sends own proposal to initiator for the security association.
- o The Initiator will start Diffie-Hellman Key Exchange Payload and Nonce payload.
- o The Responder will also start Diffie-Hellman Key Exchange Payload & Nonce payload.
- o Last two messages are encrypted used for identification & authentication of each peer.



No.	Time	Source	Destination	Protocol	Length	Info
5	22.343835	192.168.12.1	192.168.23.3	ISAKMP	206	Identity Protection (Main Mode)
6	22.437905	192.168.23.3	192.168.12.1	ISAKMP	146	Identity Protection (Main Mode)
7	22.484607	192.168.12.1	192.168.23.3	ISAKMP	326	Identity Protection (Main Mode)
8	22.578747	192.168.23.3	192.168.12.1	ISAKMP	346	Identity Protection (Main Mode)
9	22.640827	192.168.12.1	192.168.23.3	ISAKMP	142	Identity Protection (Main Mode)
10	22.750465	192.168.23.3	192.168.12.1	ISAKMP	110	Identity Protection (Main Mode)
11	22.765898	192.168.12.1	192.168.23.3	ISAKMP	214	Quick Mode

- o Initiator send odd messages and Responders send even messages.
- o Message-1, Policy proposal and attributes for Security Association & SPI Value.
- o Message-2, Policy proposal acceptance and same attributes add responder SPI Value.
- o Message-3, Diffie Helman Key exchange, Nonce (Randomly Generated Number).
- o Message-4, Diffie Helman Key exchange, Nonce number from responder.
- o Message-5, Identification and authentication packet encrypted from Initiator.
- o Message-6, Identification and authentication packet encrypted from responder.

```

> Frame 5: 206 bytes on wire (1648 bits), 206 bytes captured (1648 bits) on interface 0
> Ethernet II, Src: ca:01:03:70:00:00 (ca:01:03:70:00:00), Dst: ca:02:01:08:00:00 (ca:02:01:08:00:00)
> Internet Protocol Version 4, Src: 192.168.12.1, Dst: 192.168.23.3
> User Datagram Protocol, Src Port: 500, Dst Port: 500
< Internet Security Association and Key Management Protocol

```

Initiator SPI: c2411928cc02999f

Responder SPI: 0000000000000000

Next payload: Security Association (1)

> Version: 1.0

Exchange type: Identity Protection (Main Mode) (2)

> Flags: 0x00

Message ID: 0x00000000

Length: 164

< Payload: Security Association (1)

Next payload: Vendor ID (13)

Reserved: 00

Payload length: 56

Domain of interpretation: IPSEC (1)

> Situation: 00000001

< Payload: Proposal (2) # 1

Next payload: NONE / No Next Payload (0)

Reserved: 00

Payload length: 44

Proposal number: 1

Protocol ID: ISAKMP (1)

SPI Size: 0

Proposal transforms: 1

< Payload: Transform (3) # 1

Next payload: NONE / No Next Payload (0)

Reserved: 00

Payload length: 36

Transform number: 1

Transform ID: KEY_IKE (1)

Reserved: 0000

> IKE Attribute (t=1,l=2): Encryption-Algorithm: 3DES-CBC

> IKE Attribute (t=2,l=2): Hash-Algorithm: SHA

> IKE Attribute (t=4,l=2): Group-Description: Alternate 1024-bit MODP group

> IKE Attribute (t=3,l=2): Authentication-Method: Pre-shared key

> IKE Attribute (t=11,l=2): Life-Type: Seconds

> IKE Attribute (t=12,l=4): Life-Duration: 86400

Message-1

Proposal for ISAKMP SA

```

> Frame 6: 146 bytes on wire (1168 bits), 146 bytes captured (1168 bits) on interface 0
> Ethernet II, Src: ca:02:01:08:00:00 (ca:02:01:08:00:00), Dst: ca:01:03:70:00:00 (ca:01:03:70:00:00)
> Internet Protocol Version 4, Src: 192.168.23.3, Dst: 192.168.12.1
> User Datagram Protocol, Src Port: 500, Dst Port: 500
4 Internet Security Association and Key Management Protocol

```

```
Initiator SPI: c2411928cc02999f
```

```
Responder SPI: 66f6e3415884d2ce
```

```
Next payload: Security Association (1)
```

```
> Version: 1.0
```

```
Exchange type: Identity Protection (Main Mode) (2)
```

```
> Flags: 0x00
```

```
Message ID: 0x00000000
```

```
Length: 104
```

```
4 Payload: Security Association (1)
```

```
Next payload: Vendor ID (13)
```

```
Reserved: 00
```

```
Payload length: 56
```

```
Domain of interpretation: IPSEC (1)
```

```
> Situation: 00000001
```

```
4 Payload: Proposal (2) # 1
```

```
Next payload: NONE / No Next Payload (0)
```

```
Reserved: 00
```

```
Payload length: 44
```

```
Proposal number: 1
```

```
Protocol ID: ISAKMP (1)
```

```
SPI Size: 0
```

```
Proposal transforms: 1
```

```
4 Payload: Transform (3) # 1
```

```
Next payload: NONE / No Next Payload (0)
```

```
Reserved: 00
```

```
Payload length: 36
```

```
Transform number: 1
```

```
Transform ID: KEY_IKE (1)
```

```
Transform number: 1
```

```
Transform ID: KEY_IKE (1)
```

```
Reserved: 0000
```

```
> IKE Attribute (t=1,l=2): Encryption-Algorithm: 3DES-CBC
```

```
> IKE Attribute (t=2,l=2): Hash-Algorithm: SHA
```

```
> IKE Attribute (t=4,l=2): Group-Description: Alternate 1024-bit MODP group
```

```
> IKE Attribute (t=3,l=2): Authentication-Method: Pre-shared key
```

```
> IKE Attribute (t=11,l=2): Life-Type: Seconds
```

```
> IKE Attribute (t=12,l=4): Life-Duration: 86400
```

Message-2

Reply for the Proposal

```
4 Payload: Key Exchange (4)
```

```
Next payload: Nonce (10)
```

```
Reserved: 00
```

```
Payload length: 132
```

```
Key Exchange Data: 353be3756bebc3f85f3de91f9330f9841922c76ae1ea1178.
```

Message-3

```
4 Payload: Nonce (10)
```

```
Next payload: Vendor ID (13)
```

```
Reserved: 00
```

```
Payload length: 24
```

```
Nonce DATA: 8159202297c62812707df42f4fc141c3e8d001d3
```

Random Number

- 4 Payload: Key Exchange (4) Message-4
Next payload: Nonce (10)
Reserved: 00
Payload length: 132
Key Exchange Data: 04ca838a4bd4a875e513fa3b0ae829fcf511bf8b6550d378...
- 4 Payload: Nonce (10)
Next payload: Vendor ID (13)
Reserved: 00
Payload length: 24
Nonce DATA: 58b15e209b8ea2aefa1385e313c0d3ca14125853 Random Number

▷ Frame 9: 142 bytes on wire (1136 bits), 142 bytes captured (1136 bits) on interface 0
▷ Ethernet II, Src: ca:01:03:70:00:00 (ca:01:03:70:00:00), Dst: ca:02:01:08:00:00 (ca:02:01:08:00:00)
▷ Internet Protocol Version 4, Src: 192.168.12.1, Dst: 192.168.23.3
▷ User Datagram Protocol, Src Port: 500, Dst Port: 500
4 Internet Security Association and Key Management Protocol
Initiator SPI: c2411928cc02999f
Responder SPI: 66f6e3415884d2ce
Next payload: Identification (5) Message-5
▷ Version: 1.0
Exchange type: Identity Protection (Main Mode) (2)
▷ Flags: 0x01
Message ID: 0x00000000
Length: 100
Encrypted Data (72 bytes)

▷ Frame 10: 110 bytes on wire (880 bits), 110 bytes captured (880 bits) on interface 0
▷ Ethernet II, Src: ca:02:01:08:00:00 (ca:02:01:08:00:00), Dst: ca:01:03:70:00:00 (ca:01:03:70:00:00)
▷ Internet Protocol Version 4, Src: 192.168.23.3, Dst: 192.168.12.1
▷ User Datagram Protocol, Src Port: 500, Dst Port: 500
4 Internet Security Association and Key Management Protocol
Initiator SPI: c2411928cc02999f
Responder SPI: 66f6e3415884d2ce
Next payload: Identification (5) Message-6
▷ Version: 1.0
Exchange type: Identity Protection (Main Mode) (2)
▷ Flags: 0x01
Message ID: 0x00000000
Length: 68
Encrypted Data (40 bytes)

Aggressive Mode:

- o IKEv1 Phase 1, Aggressive mode, uses only three messages to establish IKE SA.
- o IKEv1 aggressive mode only requires three messages to establish the SA.
- o First two messages exchange include almost everything required to form IKE SA.
- o IKEv1 Phase1 Aggressive Mode is quicker than Main Mode but exchanged in cleartext.
- o First message sent from the Initiator includes SA, Proposal & Transform payload.
- o Main difference in Aggressive Mode is first message include DH Key Exchange.
- o In addition, aggressive Mode exchange the Nonce payload in first message as well.
- o Identification payload is also added in first message & sent cleartext not encrypted.
- o Responder generates Diffie-Hellman shared secret & Hash for authentication purpose.
- o The last message from the initiator is a hash that is used for authentication.

No.	Time	Source	Destination	Protocol	Length	Info
16	103.251657	ca:03:1a:dc:...	ca:03:1a:dc:...	LOOP	60	Reply
17	109.970843	192.168.12.1	192.168.23.3	ISAKMP	426	Aggressive
18	110.142297	192.168.23.3	192.168.12.1	ISAKMP	446	Aggressive
19	110.283104	192.168.12.1	192.168.23.3	ISAKMP	166	Aggressive
20	110.314587	192.168.12.1	192.168.23.3	ISAKMP	206	Quick Mode

- o **Initiator send odd messages and Responders send even message.**
- o **Message-1**, policy proposal, SA attributes, Nonce, DH and Identification.
- o **Message-2** Policy proposal, SA attributes, Nonce, DH and Hash.
- o **Message-3** from initiator is a hash value used for authentication.


```

> Internet Protocol Version 4, Src: 192.168.12.1, Dst: 192.168.23.3
> User Datagram Protocol, Src Port: 500, Dst Port: 500
4 Internet Security Association and Key Management Protocol
  Initiator SPI: 0359819557ce8d27
  Responder SPI: 0000000000000000
  Next payload: Security Association (1)
  > Version: 1.0
  Exchange type: Aggressive (4)
  > Flags: 0x00
  Message ID: 0x00000000
  Length: 384
4 Payload: Security Association (1)
  Next payload: Vendor ID (13)
  Reserved: 00
  Payload length: 56
  Domain of interpretation: IPSEC (1)
  > Situation: 00000001
4 Payload: Proposal (2) # 1
  Next payload: NONE / No Next Payload (0)
  Reserved: 00
  Payload length: 44
  Proposal number: 1
  Protocol ID: ISAKMP (1)
  SPI Size: 0
  Proposal transforms: 1
4 Payload: Transform (3) # 1
  Next payload: NONE / No Next Payload (0)
  Reserved: 00
  Payload length: 36
  Transform number: 1
  Transform ID: KEY_IKE (1)
  Reserved: 0000
  > IKE Attribute (t=1,l=2): Encryption-Algorithm: 3DES-CBC
  > IKE Attribute (t=2,l=2): Hash-Algorithm: MD5
  > IKE Attribute (t=4,l=2): Group-Description: Alternate 1024-bit MODP group
  > IKE Attribute (t=3,l=2): Authentication-Method: Pre-shared key
  > IKE Attribute (t=11,l=2): Life-Type: Seconds
  > IKE Attribute (t=12,l=4): Life-Duration: 86400
4 Payload: Key Exchange (4)
  Next payload: Nonce (10)
  Reserved: 00
  Payload length: 132
  Key Exchange Data: 127f7b17c9861458fea0895c40e389addc0556ed700c463a...
4 Payload: Nonce (10)
  Next payload: Identification (5)
  Reserved: 00
  Payload length: 24
  Nonce DATA: 21a5b1e4eef6101862e5afbd99118b14503bd272
4 Payload: Identification (5)
  Next payload: Vendor ID (13)
  Reserved: 00
  Payload length: 12
  ID type: IPV4_ADDR (1)
  Protocol ID: UDP (17)
  Port: Unused
4 Identification Data: 192.168.23.3
  ID_IPV4_ADDR: 192.168.23.3

```

Message-1

Message-1

```

> Frame 13: 446 bytes on wire (3568 bits), 446 bytes captured (3568 bits) on interface 0
> Ethernet II, Src: ca:02:01:08:00:00 (ca:02:01:08:00:00), Dst: ca:01:03:70:00:00 (ca:01:03:70:00:00)
> Internet Protocol Version 4, Src: 192.168.23.3, Dst: 192.168.12.1
> User Datagram Protocol, Src Port: 500, Dst Port: 500
4 Internet Security Association and Key Management Protocol

```

```

    Initiator SPI: 0359819557ce8d27

```

```

    Responder SPI: 22247c3a711060a7

```

```

    Next payload: Security Association (1)

```

Message-2

```

> Version: 1.0

```

```

    Exchange type: Aggressive (4)

```

```

> Flags: 0x00

```

```

    Message ID: 0x00000000

```

```

    Length: 404

```

```

4 Payload: Security Association (1)

```

```

    Next payload: Vendor ID (13)

```

```

    Reserved: 00

```

```

    Payload length: 56

```

```

    Domain of interpretation: IPSEC (1)

```

```

> Situation: 00000001

```

```

4 Payload: Proposal (2) # 1

```

```

    Next payload: NONE / No Next Payload (0)

```

```

    Reserved: 00

```

```

    Payload length: 44

```

```

    Proposal number: 1

```

```

    Protocol ID: ISAKMP (1)

```

```

    SPI Size: 0

```

```

    Proposal transforms: 1

```

```

4 Payload: Transform (3) # 1

```

```

    Next payload: NONE / No Next Payload (0)

```

```

    Reserved: 00

```

```

    Payload length: 36

```

```

    Transform number: 1

```

```

    Transform ID: KEY_IKE (1)

```

```

    Reserved: 0000

```

```

> IKE Attribute (t=1,l=2): Encryption-Algorithm: 3DES-CBC

```

```

> IKE Attribute (t=2,l=2): Hash-Algorithm: MD5

```

```

> IKE Attribute (t=4,l=2): Group-Description: Alternate 1024-bit MODP group

```

```

> IKE Attribute (t=3,l=2): Authentication-Method: Pre-shared key

```

```

> IKE Attribute (t=11,l=2): Life-Type: Seconds

```

```

> IKE Attribute (t=12,l=4): Life-Duration: 86400

```


4 Payload: Key Exchange (4)
Next payload: Identification (5)
Reserved: 00
Payload length: 132
Key Exchange Data: 13ecab9f32fa63fe76c187ac5299b8d2fbb1c9bd1c6484c5...

Message-2

4 Payload: Identification (5)
Next payload: Nonce (10)
Reserved: 00
Payload length: 12
ID type: IPV4_ADDR (1)
Protocol ID: Unused
Port: Unused

4 Identification Data: 192.168.23.3
ID_IPV4_ADDR: 192.168.23.3

4 Payload: Nonce (10)
Next payload: Hash (8)
Reserved: 00
Payload length: 24
Nonce DATA: 5d0a80694262d86f9e3d8ef6cc8c1cf7b8393cf7

4 Payload: Hash (8)
Next payload: NAT-D (RFC 3947) (20)
Reserved: 00
Payload length: 20
Hash DATA: 1cdb59f1b4978d0d3c8ea30c3f0771b8

- ▷ Frame 14: 166 bytes on wire (1328 bits), 166 bytes captured (1328 bits) on interface 0
- ▷ Ethernet II, Src: ca:01:03:70:00:00 (ca:01:03:70:00:00), Dst: ca:02:01:08:00:00 (ca:02:01:08:00:00)
- ▷ Internet Protocol Version 4, Src: 192.168.12.1, Dst: 192.168.23.3
- ▷ User Datagram Protocol, Src Port: 500, Dst Port: 500
- 4 Internet Security Association and Key Management Protocol

Message-3

Initiator SPI: 0359819557ce8d27
Responder SPI: 22247c3a711060a7
Next payload: Hash (8)

- ▷ Version: 1.0
- Exchange type: Aggressive (4)
- ▷ Flags: 0x01
- Message ID: 0x00000000
- Length: 124
- Encrypted Data (96 bytes)

IKEv1 Phase 2:

- o IKE Phase 2 Tunnel (IPsec Tunnel) actually used to protect user data.
- o IKEv2 Phase 2 negotiation is done in only one mode that is **Quick Mode**.
- o IKEv1 Phase 2 (Quick Mode) consists of three 3-message exchanges.
- o Message exchanges in Phase 2 are protected by encryption & authentication.
- o IKEv1 Phase 2 Quick Mode using the keys derived in the IKEv1 Phase 1.
- o When IKEv1 phase1 is successfully completed, then Phase 2 is started.
- o If IKEv1 phase 1 is not successfully, completed IKEv1 Phase 2 will not start.
- o Just like in IKE phase 1, the peers will negotiate about a number of items.
- o Negotiate about **Encryption Algorithm** (DES, 3DES or AES different bits).
- o Negotiate about **Authentication or Hashing Algorithm** (MD5, SHA-1 or SHA-2).
- o Negotiate about **IPSec Protocol Encapsulation Protocol** (AH or ESP).
- o Negotiate about **SA lifetime** (Time in seconds or data transfer in kilobytes).
- o Negotiate about **Encapsulation Mode** (Tunnel Mode or Transport Mode).

```

> Frame 15: 206 bytes on wire (1648 bits), 206 bytes captured (1648 bits) on interface 0
> Ethernet II, Src: ca:01:03:70:00:00 (ca:01:03:70:00:00), Dst: ca:02:01:08:00:00 (ca:02:01:08:00:00)
> Internet Protocol Version 4, Src: 192.168.12.1, Dst: 192.168.23.3
> User Datagram Protocol, Src Port: 500, Dst Port: 500
4 Internet Security Association and Key Management Protocol
  Initiator SPI: 0359819557ce8d27
  Responder SPI: 22247c3a711060a7
  Next payload: Hash (8)
  > Version: 1.0
  Exchange type: Quick Mode (32)
  > Flags: 0x01
  Message ID: 0x90928e08
  Length: 164
  Encrypted Data (136 bytes)
```

Message-1

Indicate Phase 2

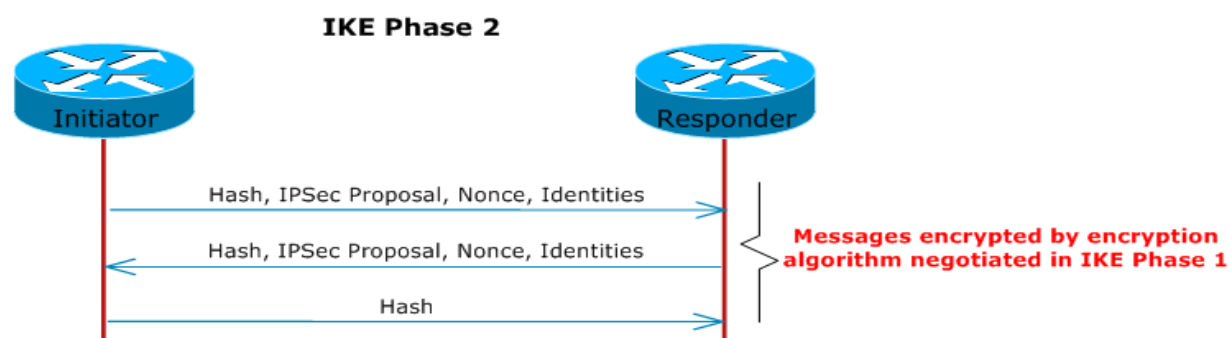
Encrypted Data

```

> Frame 16: 206 bytes on wire (1648 bits), 206 bytes captured (1648 bits) on interface 0
> Ethernet II, Src: ca:02:01:08:00:00 (ca:02:01:08:00:00), Dst: ca:01:03:70:00:00 (ca:01:03:70:00:00)
> Internet Protocol Version 4, Src: 192.168.23.3, Dst: 192.168.12.1
> User Datagram Protocol, Src Port: 500, Dst Port: 500
4 Internet Security Association and Key Management Protocol
  Initiator SPI: 0359819557ce8d27
  Responder SPI: 22247c3a711060a7
  Next payload: Hash (8)
  > Version: 1.0
  Exchange type: Quick Mode (32)
  > Flags: 0x01
  Message ID: 0x90928e08
  Length: 164
  Encrypted Data (136 bytes)
```

Message-2

▶ Frame 17: 94 bytes on wire (752 bits), 94 bytes captured (752 bits) on interface 0
 ▶ Ethernet II, Src: ca:01:03:70:00:00 (ca:01:03:70:00:00), Dst: ca:02:01:08:00:00 (ca:02:01:08:00:00)
 ▶ Internet Protocol Version 4, Src: 192.168.12.1, Dst: 192.168.23.3
 ▶ User Datagram Protocol, Src Port: 500, Dst Port: 500
 4 Internet Security Association and Key Management Protocol **Message-3**
 Initiator SPI: 0359819557ce8d27
 Responder SPI: 22247c3a711060a7
 Next payload: Hash (8)
 ▶ Version: 1.0
 Exchange type: Quick Mode (32)
 ▶ Flags: 0x01
 Message ID: 0x90928e08
 Length: 52
 Encrypted Data (24 bytes)



IKEv2 Phase Mode:

- o IKEv2 does not have main or aggressive mode for phase 1 & no quick mode in phase 2.
- o Internet Key Exchange (IKEv2) also has two Phase negotiation process.
- o First Phase is known as **IKE_SA_INIT** and the second Phase is called as **IKE_AUTH**.
- o In Internet Key Exchange (IKEv2), only four messages are required for entire exchange.
- o Internet Key Exchange (IKEv2) runs over UDP ports 500 and 4500 (IPsec NAT Traversal).
- o Devices configured to use IKEv2 accept packets from UDP ports 500 and 4500.
- o IKEv2 IPsec Peers can be validated using Pre-Shared Keys, Certificates, or EAP.
- o In IKEv2, the First message from Initiator to Responder (IKE_SA_INIT).
- o Contains SA proposals, Encryption & Integrity algorithms, DH keys & Nonce.
- o In IKEv2, second message from Responder to Initiator (IKE_SA_INIT).
- o Contains SA proposals, Encryption & Integrity algorithms, DH keys & Nonce.
- o After Messages 1 & 2, next messages are protected by encrypting & authenticating.
- o Third and fourth messages (IKE_AUTH) are encrypted and authenticated.
- o Over the IKE SA created by the previous Messages 1 and 2 (IKE_SA_INIT).
- o IKEv2 these two messages are for Authentication.
- o Initiators and responder's identity, certificates exchange is completed at this stage.
- o Third & fourth messages (IKE_AUTH) are used authenticate the previous messages.
- o Validate the identity of IPsec peers and to establish the first CHILD_SA.
- o IKEv2, there is one tunnel for the control channel called "IKE Tunnel".
- o IKEv2, the second tunnel for the user traffic called "Child Tunnel"
- o "IKE Tunnel" uses UDP port 500 for negotiation while "IPsec Tunnel" uses ESP port 50.

No.	Time	Source	Destination	Protocol	Length	Info
941	3848.024464	192.168.12.1	192.168.23.2	ISAKMP	314	IKE_SA_INIT MID=00 Initiator Request
942	3848.127723	192.168.23.2	192.168.12.1	ISAKMP	314	IKE_SA_INIT MID=00 Responder Response
943	3848.189471	192.168.12.1	192.168.23.2	ISAKMP	346	IKE_AUTH MID=01 Initiator Request
944	3848.237781	192.168.23.2	192.168.12.1	ISAKMP	314	IKE_AUTH MID=01 Responder Response

First Message from Initiator to Responder (IKE_SA_INIT) contains the SA proposals, Encryption and Integrity algorithms, Diffie-Hellman keys and Nonce.

Second Message from Responder to Initiator (IKE_SA_INIT) contains the SA proposals, Encryption and Integrity algorithms, Diffie-Hellman keys and Nonce.

Third and Fourth Messages (IKE_AUTH) are encrypted and authenticated over the IKE SA created by the previous Messages 1 and 2 (IKE_SA_INIT). These two messages are for Authentication. Initiator's and responder's identity, certificates exchange (if available) are completed at this stage. The third and fourth messages (IKE_AUTH) are used authenticate the previous messages, validate the identity of IPsec peers and to establish the first CHILD_SA. At the end of messages 3 & 4, identities of IPsec Peers are verified and first CHILD_SA is established.

․ Frame 941: 314 bytes on wire (2512 bits), 314 bytes captured (2512 bits) on interface 0
 ․ Ethernet II, Src: ca:01:0e:5c:00:00 (ca:01:0e:5c:00:00), Dst: ca:02:08:3c:00:00 (ca:02:08:3c:00:00)
 ․ Internet Protocol Version 4, Src: 192.168.12.1, Dst: 192.168.23.2
 ․ User Datagram Protocol, Src Port: 500, Dst Port: 500
 4 Internet Security Association and Key Management Protocol

Message-1

Initiator SPI: 495002db8956cfd0
 Responder SPI: 0000000000000000
 Next payload: Security Association (33)
 ․ Version: 2.0
 Exchange type: IKE_SA_INIT (34)
 ․ Flags: 0x08 (Initiator, No higher version, Request)
 Message ID: 0x00000000
 Length: 272
 ․ Payload: Security Association (33)
 ․ Payload: Key Exchange (34)
 ․ Payload: Nonce (40)
 ․ Payload: Vendor ID (43) : Cisco Delete Reason Supported
 ․ Payload: Vendor ID (43) : Cisco FlexVPN Supported
 ․ Payload: Notify (41) - NAT_DETECTION_SOURCE_IP
 ․ Payload: Notify (41) - NAT_DETECTION_DESTINATION_IP

․ Frame 942: 314 bytes on wire (2512 bits), 314 bytes captured (2512 bits) on interface 0
 ․ Ethernet II, Src: ca:02:08:3c:00:00 (ca:02:08:3c:00:00), Dst: ca:01:0e:5c:00:00 (ca:01:0e:5c:00:00)
 ․ Internet Protocol Version 4, Src: 192.168.23.2, Dst: 192.168.12.1
 ․ User Datagram Protocol, Src Port: 500, Dst Port: 500
 4 Internet Security Association and Key Management Protocol

Message-2

Initiator SPI: 495002db8956cfd0
 Responder SPI: 381e9a32bca4337f
 Next payload: Security Association (33)
 ․ Version: 2.0
 Exchange type: IKE_SA_INIT (34)
 ․ Flags: 0x20 (Responder, No higher version, Response)
 Message ID: 0x00000000
 Length: 272
 ․ Payload: Security Association (33)
 ․ Payload: Key Exchange (34)
 ․ Payload: Nonce (40)
 ․ Payload: Vendor ID (43) : Cisco Delete Reason Supported
 ․ Payload: Vendor ID (43) : Cisco FlexVPN Supported
 ․ Payload: Notify (41) - NAT_DETECTION_SOURCE_IP
 ․ Payload: Notify (41) - NAT_DETECTION_DESTINATION_IP

․ Frame 943: 346 bytes on wire (2768 bits), 346 bytes captured (2768 bits) on interface 0
 ․ Ethernet II, Src: ca:01:0e:5c:00:00 (ca:01:0e:5c:00:00), Dst: ca:02:08:3c:00:00 (ca:02:08:3c:00:00)
 ․ Internet Protocol Version 4, Src: 192.168.12.1, Dst: 192.168.23.2
 ․ User Datagram Protocol, Src Port: 500, Dst Port: 500
 4 Internet Security Association and Key Management Protocol

Message-3

Initiator SPI: 495002db8956cfd0
 Responder SPI: 381e9a32bca4337f
 Next payload: Encrypted and Authenticated (46)
 ․ Version: 2.0
 Exchange type: IKE_AUTH (35)
 ․ Flags: 0x08 (Initiator, No higher version, Request)
 Message ID: 0x00000001
 Length: 304
 ․ Payload: Encrypted and Authenticated (46)

▷ Frame 944: 314 bytes on wire (2512 bits), 314 bytes captured (2512 bits) on interface 0
 ▷ Ethernet II, Src: ca:02:08:3c:00:00 (ca:02:08:3c:00:00), Dst: ca:01:0e:5c:00:00 (ca:01:0e:5c:00:00)
 ▷ Internet Protocol Version 4, Src: 192.168.23.2, Dst: 192.168.12.1
 ▷ User Datagram Protocol, Src Port: 500, Dst Port: 500
 4 Internet Security Association and Key Management Protocol

Message-4

Initiator SPI: 495002db8956cfd0
 Responder SPI: 381e9a32bca4337f
 Next payload: Encrypted and Authenticated (46)
 ▷ Version: 2.0
 Exchange type: IKE_AUTH (35)
 ▷ Flags: 0x20 (Responder, No higher version, Response)
 Message ID: 0x00000001
 Length: 272
 ▷ Payload: Encrypted and Authenticated (46)

